# PHASE 8: SOFTWARE MAINTENANCE AND PROFESSIONAL, LEGAL AND ETHICAL ISSUES

## 8.1 Introduction

This phase discusses the maintenance of the Learning Management System (LMS), focusing on an improvement identified after the release of version 1.0. It also considers the professional, legal and ethical responsibilities involved in maintaining the system.

## 8.2 Maintenance Request (CR1)

The identified change request is to allow lecturers to extend an assignment deadline for the whole class. The updated deadline must also be displayed to students when they view their assignments.

### 8.2.1 Type of Maintenance

This is **perfective maintenance** because it improves the existing system by adding functionality that makes assignment management more flexible.

The feature was not included in version 1.0 and was deferred for future development.

### 8.2.2 Impact Analysis

| System area | Required change |
|---|---|
| Requirements | Add a requirement allowing lecturers to extend deadlines for the whole class. |
| Design | Update the use-case, class and sequence diagrams to include class-wide extensions. |
| Classes and components | Update AssignmentService and the relevant repository components to manage extensions. |
| Database | Add a `class_extensions` table to store extension records. |
| Testing | Add tests for lecturer permissions, deadline updates and deadline calculations. |

### 8.2.3 Proposed Implementation

A new `class_extensions` table will store the assignment ID, new deadline, reason, lecturer who created the extension and creation date.

The system will determine the effective deadline using the following order:

1. An individual student's extension, if one exists.
2. The latest class-wide extension, if one exists.
3. The original assignment deadline if no extension exists.

Only authorised lecturers will be allowed to create class-wide extensions. The updated deadline will be displayed to students.

### 8.2.4 Regression Testing

The following tests are planned before releasing the maintenance update:

| Test | Expected result |
|---|---|
| Lecturer extends a class deadline | New deadline is saved successfully. |
| Student views the assignment | Updated deadline is displayed. |
| Student has an individual extension | Individual deadline takes priority. |
| No extension exists | Original deadline is used. |
| Unauthorised user attempts an extension | Access is denied. |
| Existing submission and marking functions are tested | They continue working correctly. |

Version 1.0 currently has 104 automated tests, all of which passed in the latest test run. These tests provide the regression baseline. The additional CR1 tests will be executed after implementation.

### 8.2.5 Planned Maintenance Release

After implementation and successful testing, the changes will be included in version 1.1. The updated version will be built, tested and released with documentation describing the new functionality.

Version 1.1 has not yet been released.

## 8.3 Professional, Legal and Ethical Issues

### 8.3.1 Legal Responsibilities

The system handles personal information and academic records. It must therefore be managed in accordance with applicable data-protection requirements in Lesotho, including the Data Protection Act, 2011 (Act No. 5 of 2012). Personal information should only be accessed and used for legitimate academic purposes.

### 8.3.2 Security and Privacy

The system uses password hashing and role-based access control to protect user accounts and academic information. Developers and administrators must protect assessment records from unauthorised access, disclosure or modification. Database backups should also be stored securely.

### 8.3.3 Accessibility

The system should be easy to understand and use by all authorised users. Clear instructions, readable messages and simple navigation should be maintained to reduce difficulties when using the system.

### 8.3.4 Ethical Responsibilities

The system must treat students fairly. Late penalties and deadline extensions should be applied consistently, while marking and moderation decisions should remain transparent and accountable. User information must not be misused or shared without proper authorisation.

### 8.3.5 Professional Responsibilities

Developers should maintain proper documentation, test changes before release, correct identified defects and follow good software engineering practices. Changes should also be reviewed to avoid introducing new errors or affecting existing functionality.

## 8.4 Conclusion

Phase 8 identifies a planned improvement to the LMS through class-wide assignment deadline extensions. The proposed changes include database updates, service modifications and regression testing before version 1.1 is released. The phase also highlights the legal, professional and ethical responsibilities required to maintain a secure, fair and reliable system.